



Airline Contrail Impact Report 2024

What are Contrails?

Contrails warm the planet because contrail clouds act like a blanket on Earth and have a net heating effect. The 2022 IPCC report noted that clouds created by contrails account for roughly 35% of aviation's global warming impact.

What is Global Warming Potential (GWP)?

GWP measures how much warming contrails cause over a number of years compared to CO2. Contrails heat the Earth quickly but for a short time. To align with the guidelines from the EU Non-CO2 MRV report the contrail impact is shown in CO2e over 20, 50 and 100 years. The middle value, GWP50, is used as default.

Methodology, data sources & assumptions

IATA Codes: This report covers [] IATA code.

Modeling Process:

- Flight trajectories are built from ADS-B data and undergo quality control, removing bad trajectories (this reduces the number of analyzed flights).
- The CoCiP physics-based model predicts contrails and CO2e impact.
- Aircraft type (e.g., B777) is used to select a default engine type for modeling, unless the airline provides specific aircraft-engine pairings.

Fuel Consumption Estimates:

- Fuel burn is based on the physics model, sensitive to engine type and assumed constant load factors (per aircraft type).
- It primarily reflects fuel consumption "during flight" or "at cruise," generally excluding taxi, and sometimes take-off/landing.

Data Discrepancies:

- The report uses modeled estimates. An airline's internal data regarding fuel use, load factor, exact trajectory, engine configuration, etc., is more precise.
- Discrepancies between the report's modeled numbers and the airline's internal figures are expected.

Report Exclusions:

- Excludes aircraft types that don't produce contrails, or flights with insufficient or corrupt data.
- Flights are excluded if the aircraft type isn't in the database (mainly affecting propeller or some private aircraft).

Impact Data

Based on our prediction model, we analyzed 1.4M flights, 5.5 Million km (55,501 flight hours) and of those, 4.4% of all test flights generated warming contrails in 2024.

Contrails percentage	test Average CO2e/km flown	All Airlines average CO2e/km	Average fuel burn
4.4% of flights km	20.9 kg CO2e/km	23.4 kg CO2e/km	26.6 kg CO2e/km

Global Warming potential 100, 50, 20
Contrails are more warming in the short term

